14. $\int_1^\infty \frac{e^{-1/x}}{x^2}\,dx$

Let $I(t) = \int_1^t \frac{1}{x^2} e^{-1/x}\,dx$

$t > 1$

$= \int_{-1/t}^{-1} e^u\,du = \left\{ e^{-1} - e^{-1/t} \right\}$

$u = -\frac{1}{x}$

$du = \frac{1}{x^2}\,dx$

Then $I_1(t) \to e^{-1}$ as $t \to \infty$, so

$$\int_1^\infty \frac{e^{-1/x}}{x^2}\,dx = e^{-1}$$

31. $\int_{-2}^3 \frac{1}{x^4}\,dx$

$I_1(t) = \int_{-2}^t \frac{1}{x^4}\,dx$

$-2 < t < 0$

$= -\frac{1}{3x^3}\Big|_{-2}^t = -\frac{1}{3}\left\{ \frac{1}{t^3} + \frac{1}{2} \right\} \to +\infty$ as $t \to 0^-$

Thus the integral DNE.

50. $\int_1^\infty \frac{1+\sin^2 x}{\sqrt{x}}\,dx$ $\leftarrow$ Does this converge / diverge?

Use the comparison test.

Observe that $0 \le \sin^2 x \le 1$ for all x, so

$$0 \le \frac{1}{\sqrt{x}} \le \frac{1+\sin^2 x}{\sqrt{x}} \le \frac{2}{\sqrt{x}} \quad \text{for all } x \ge 1.$$

Now $\displaystyle\int_1^\infty \frac{1}{\sqrt{x}}\,dx = 2\lim_{t\to\infty}\left(t^{1/2} - 1\right) = \infty$,

so the integral <u>diverges</u> to ∞.